

Name: _____

Rollo: _____

Research Methodology
MS(AI), Dr. Zohair Ahmed
Activity Sheet 1: Literature Review & Journal Quality

Part A – Searching Papers

1. Use **Google Scholar** to search for:
 - "cyberbullying AND deep learning"
 - Write down the **first 3 paper titles** you find.
 - Which years were they published?

2. Use **Scopus**
(<https://www.scopus.com/sources>):
 - Search for "blockchain AND energy management".
 - Apply a filter: **PUBYEAR > 2021**.
 - How many results appear?

Part B – Using Keywords & Operators

Fill in the missing operator:

1. "machine learning ____ natural language processing" (narrow the search).
2. "hate speech ____ cyberbullying" (broaden the search).
3. "neural networks ____ reinforcement" (exclude reinforcement).

Wildcard practice:

- What results might you get from searching **comput*?**
(Write at least 3 variations):

Part C – Journals vs. Conferences

Match the features to **Journal (J)** or **Conference (C)**:

1. Faster publication, early-stage work → ____
2. More detailed, rigorous review → ____
3. Networking and presentation opportunities → ____
4. Indexed in WoS/Scopus with Impact Factor → ____

Part D – Indexing Systems

Fill in the blanks:

1. **WoS (Clarivate)** includes indexes such as _____, _____, _____.
2. **Scopus** is maintained by _____.
3. **Quartile rankings (Q1–Q4)** are given by (WoS / Scopus)? _____.

Part F – Peer Review Process

Arrange the steps in correct order (number them 1–6):

- ____ Author revises and resubmits.
- ____ Reviewers recommend accept/reject/revise.
- ____ Paper submitted to journal.
- ____ Final decision by editor.
- ____ Paper sent to reviewers.
- ____ Accepted paper published.

Activity Sheet 2 – Research Problem Identification

Part A – Choosing a Broad Area

1. Write down **three broad research areas** you are interested in (e.g., NLP, Cybersecurity, Healthcare AI, Blockchain).
 - Area 1: _____
 - Area 2: _____
 - Area 3: _____

Part B – Finding Gaps

1. Search for **two recent papers (2020–2025)** in your chosen area.
 - Paper 1: Title, Year, Journal/Conference

 - Paper 2: Title, Year, Journal/Conference
2. From each paper, note one **limitation or gap** mentioned by the authors:
 - Gap from Paper 1

 - Gap from Paper

Part C – Converting Gap into Research Problem

1. Choose one gap you found above.
2. Write a **research problem statement** based on that gap:
 - Gap:

 - Problem Statement:

Part D – Problem vs. Gap Practice

Fill in the blanks:

1. **Gap:** Few studies on _____.
Problem: How can _____ be applied to solve _____?
2. **Gap:** Existing methods have low accuracy in _____.
Problem: Designing a _____ approach to improve _____.

Part E – Reflection

- Why is it important to spend time on **problem finding** instead of jumping straight to solutions?
(Write 3–4 sentences)